

Part 1 – Project Objective

Our objective: To increase the number of rentals in the second quarter (Q2) of 2022, and to build an action plan based on the company's data.

We started by writing queries to understand what is happening in the business:

1) Cleaning up anomalous trip errors:

A query that identifies system issues. For example: trips that are too short (less than 2 minutes), which usually indicates faulty bikes; or trips that are too long (over 24 hours), which indicates bikes that were stolen or forgotten.

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	ride_status	total_trips	percentage_of_to...		
1	Valid Ride	16579	99.96		
2	System Error: Stolen / Forgotte...	6	0.04		

2) Peak times – when do people ride?:

A query that shows which months, days, and hours are the most popular, and how long an average trip lasts.

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	month	day_of_week	hour_of_day	total_rentals	avg_duration_min...
1		3. Tuesday	15	149	17.9
2		3. Tuesday	17	128	10.2
3		2. Monday	16	128	26.7
4		3. Tuesday	15	126	9.8
5		4. Wednesday	12	126	12.9
6		3. Tuesday	18	125	39.9
7		2. Monday	17	122	26.1
8		5. Thursday	15	120	6.9
9		5. Thursday	17	120	21.1
10		3. Tuesday	18	119	17.8
11		3. Tuesday	13	116	10.6
12		3. Tuesday	12	116	9.9
13		2. Monday	13	109	9.9
14		4. Wednesday	15	109	10.0

3) Popular routes:

Code that finds which origin (start) and destination (end) stations are the most in-demand, in order to identify the customers' main "traffic arteries" (core transit routes).

Our results:

Job information		Results	Visualization	JSON	Execution details	Exec
Row	station_name	total_departures	total_arrivals	total_station_acti...		
1	Dean Keeton/Speedway	3001	1149	4150		
2	21st/Speedway @ PCL	0	3552	3552		
3	21st/Guadalupe	2298	916	3214		
4	23rd/San Gabriel	2111	874	2985		
5	22nd/Pearl	1835	1006	2841		
6	28th/Rio Grande	1857	917	2774		

4. Negative Station Analysis

This query identified the stations with the highest negative net bike flow, meaning stations where bikes leave faster than they arrive. These stations experience high demand and are at greater risk of running out of available bikes.

The results helped us identify critical locations where bike shortages are most likely to occur and where rebalancing efforts should be prioritized to maintain service availability.

Row	station_name	total_departures	total_arrivals	net_flow		
1	Dean Keeton/Speedway	3001	1149	-1852		
2	21st/Guadalupe	2298	916	-1382		
3	23rd/San Gabriel	2111	874	-1237		
4	28th/Rio Grande	1857	917	-940		
5	21st/University	1578	732	-846		
6	22nd/Pearl	1835	1006	-829		

5. Station Balance Analysis

This query identified both stations with a negative net bike flow and stations that are close to becoming negative. These are stations where departures are equal to or slightly higher than arrivals, indicating a risk of future bike shortages.

The analysis helped detect stations that are already experiencing inventory problems, as well as stations that may become problematic if current demand patterns continue. This information supports proactive rebalancing and inventory management decisions.

Row	station_name	total_departures	total_arrivals	net_flow	station_status
1	Dean Keeton/Speedway	3001	1149	-1852	Critically Draining (High Deman...
2	21st/Guadalupe	2298	916	-1382	Critically Draining (High Deman...
3	23rd/San Gabriel	2111	874	-1237	Critically Draining (High Deman...
4	28th/Rio Grande	1857	917	-940	Critically Draining (High Deman...
5	21st/University	1578	732	-846	Critically Draining (High Deman...
6	22nd/Pearl	1835	1006	-829	Critically Draining (High Deman...
7	23rd/San Jacinto @ DKR Stadium	1378	573	-805	Critically Draining (High Deman...
8	Electric Drive/Sandra Muraida ...	822	174	-648	Critically Draining (High Deman...
9	East 6th/Medina	731	107	-624	Critically Draining (High Deman...
10	22.5/Rio Grande	968	403	-565	Critically Draining (High Deman...
11	Springfest 2022	0	1	1	Leaning Toward Surplus
12	Rio Grande/12th	0	6	6	Leaning Toward Surplus
13	East 2nd/Pedernales	0	7	7	Leaning Toward Surplus
14	Rainey/Davis	0	8	8	Leaning Toward Surplus

6. Overflowed Station Analysis

This query identified the stations with the highest positive net bike flow, meaning stations where more bikes arrive than depart.

These stations are at risk of becoming overcrowded, which may result in a shortage of available docking spaces for returning bikes. The analysis helped identify locations where bike accumulation is most severe and where redistribution efforts are needed to prevent congestion and maintain smooth network operations.

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	station_name	total_departures	total_arrivals	net_flow	station_status
1	21st/Speedway @ PCL	0	3552	3552	Critically Overflowing (High Ret...
2	26th/Nueces	0	1379	1379	Critically Overflowing (High Ret...
3	Dean Keeton/Whitis	0	1099	1099	Critically Overflowing (High Ret...
4	Guadalupe/West Mall @ Univer...	0	1007	1007	Critically Overflowing (High Ret...
5	3rd/West	0	137	137	Critically Overflowing (High Ret...
6	16th/San Antonio	0	103	103	Critically Overflowing (High Ret...
7	3rd/Nueces	0	96	96	Critically Overflowing (High Ret...
8	Riverside/South Lamar	0	92	92	Critically Overflowing (High Ret...
9	9th/Henderson	0	82	82	Critically Overflowing (High Ret...
10	11th/Congress @ The Texas Ca...	0	74	74	Critically Overflowing (High Ret...
11	4th/Guadalupe @ Republic Squ...	0	74	74	Critically Overflowing (High Ret...
12	6th/Congress	0	73	73	Critically Overflowing (High Ret...
13	5th/Bowie	0	73	73	Critically Overflowing (High Ret...
14	6th/Brazos	0	71	71	Critically Overflowing (High Ret...

This query identified the stations with the highest positive net bike flow. These are stations where many bikes are returned, but very few bikes are taken out again.

This is important because these stations may become overcrowded, leaving customers without available docking spaces at the end of their ride. This can increase operational pressure on the company, require more frequent bike redistribution, and may even lead to lost or improperly returned bikes.

The results suggest that these stations should be prioritized in Q2 by either adding more physical docking spaces or encouraging customers to return bikes to nearby stations with lower inventory.

This finding also led us to an important follow-up question:

Is there a relationship between the negative stations, where bikes are taken from but not returned, and the positive stations, where bikes are returned but not taken out again?

To answer this, we decided to analyze the routes between negative and positive stations and check whether customers are moving bikes from one type of station to another.

7. Route Analysis Between Negative and Positive Stations

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	start_station	start_station_net...	end_station	end_station_net...	total_trips_on_ro...
1	Dean Keeton/Speedway	-1852	21st/Speedway @ PCL	3552	831
2	23rd/San Gabriel	-1237	21st/Speedway @ PCL	3552	699
3	22nd/Pearl	-829	21st/Speedway @ PCL	3552	596
4	Dean Keeton/Speedway	-1852	26th/Nueces	1379	426
5	21st/Guadalupe	-1382	21st/Speedway @ PCL	3552	415
6	28th/Rio Grande	-940	21st/Speedway @ PCL	3552	393
7	21st/Guadalupe	-1382	26th/Nueces	1379	316
8	22.5/Rio Grande	-565	21st/Speedway @ PCL	3552	296
9	Dean Keeton/Speedway	-1852	Dean Keeton/Whitis	1099	267
10	21st/Guadalupe	-1382	22nd/Pearl	-829	253
11	28th/Rio Grande	-940	28th/Rio Grande	-940	251
12	23rd/San Jacinto @ DKR Stadium	-805	21st/Speedway @ PCL	3552	233
13	21st/University	-846	26th/Nueces	1379	232
14	23rd/San Gabriel	-1237	Guadalupe/West Mall @ Univer...	1007	230

We then analyzed the routes connecting negative stations to positive stations in order to understand where all of these bikes were coming from.

The results showed that a large number of bikes arriving at PCL originated from stations with severe negative balances. For example:

- Route 1 (831 trips) originated from **Dean Keeton/Speedway**, which had a net bike flow of **-1,852**.

- Route 2 (699 trips) originated from **23rd/San Gabriel**, which had a net bike flow of **-1,237**.
- Route 3 (596 trips) originated from **22nd/Pearl**, which had a net bike flow of **-829**.

These stations are located in residential and student housing areas around West Campus. Every morning, students leave their homes, take bikes from these stations, and ride to the Perry-Castañeda Library (PCL). As a result, the residential stations become depleted while PCL continuously accumulates bikes.

PCL @ 21st/Speedway can be described as the network's **"bottomless pit."** Nearly all of the most popular routes during Q1 ended at this station. Its net bike flow reached **+3,552**, meaning that thousands of excess bikes accumulated there with very little outbound demand.

This creates a significant operational challenge, as the station is constantly at risk of running out of available docking spaces. PCL is located directly outside the Perry-Castañeda Library on the University of Texas campus, making it a major destination hub for students and faculty during the morning hours.

This analysis revealed the geographic root cause of the imbalance problem and explained why certain stations consistently lose bikes while others continuously accumulate them.

8. Customer Segment Analysis

Query results						Chat	Save results
Job information		Results	Visualization	JSON	Execution details	Execution graph	
Row	customer_segment	total_rentals	total_minutes_util...	avg Ride duration...	share_of_total_re...		
1	Student Membership	12595	196106	15.6	75.97		
2	Local31	1310	44675	34.1	7.9		
3	Local365	1074	22785	21.2	6.48		
4	Pay-as-you-ride	444	16281	36.7	2.68		
5	Explorer	406	20462	50.4	2.45		
6	3-Day Weekender	240	12037	50.2	1.45		
7	U.T. Student Membership	223	3828	17.2	1.35		
8	24 Hour Walk Up Pass	143	8749	61.2	0.86		
9	Single Trip (Pay-as-you-ride)	138	7048	51.1	0.83		
10	HT Ram Membership	4	295	73.8	0.02		

After identifying the most problematic stations in the network, we wanted to investigate whether there was a relationship between the company's most valuable customer groups and the stations experiencing the greatest imbalances. Therefore, we first analyzed the customer segments to understand which groups generate the majority of the company's activity.

Student Members – The Core Volume Driver

The most significant finding was that Student Membership and U.T. Student Membership users together accounted for approximately 77.3% of all rentals (12,595 trips).

Their average ride duration was relatively short, at approximately 15.6 minutes per trip.

This finding strongly supports our previous conclusion. Students typically pick up bikes from residential areas in the morning, ride for about 15 minutes to campus, and return the bikes near the Perry-Castañeda Library (PCL). As a result, Zen City's business is heavily dependent on student commuting patterns.

Local Members – The Stable Customer Base

The Local31 and Local365 membership plans together accounted for approximately 14.4% of all rentals.

In particular, Local31 users had an average ride duration of 34.1 minutes, which is nearly twice the average duration of student riders. This suggests that local members use the bikes more extensively throughout the city and contribute to steady, recurring demand.

Casual Riders – The Highest Revenue Potential per Ride

Customer groups such as Explorer, 3-Day Weekender, and 24-Hour Walk-Up Pass represented only a small share of total rentals, typically around 1–2% each.

However, their average ride durations ranged from 50 to 61 minutes per trip, significantly longer than any other customer segment.

From a business perspective, these riders are likely to generate higher revenue per trip because they typically pay premium rates for longer rides. Although they account for a smaller share of total rentals, they represent an important source of profitability and should not be overlooked in future growth strategies.

9. Trip Duration Analysis for Critical Stations

Row	station_group	station_name	net_flow	measured_metric	total_trips	avg_duration_min...
1	Top 5 Positive (Overflowing Stat...	3rd/West	137	Arrivals (Trips Ending Here)	137	21.4
2	Top 5 Positive (Overflowing Stat...	Guadalupe/West Mall @ Univer...	1007	Arrivals (Trips Ending Here)	1007	10.7
3	Top 5 Positive (Overflowing Stat...	Dean Keeton/Whitis	1099	Arrivals (Trips Ending Here)	1099	24.0
4	Top 5 Positive (Overflowing Stat...	26th/Nueces	1379	Arrivals (Trips Ending Here)	1379	16.0
5	Top 5 Positive (Overflowing Stat...	21st/Speedway @ PCL	3552	Arrivals (Trips Ending Here)	3552	9.0
6	Top 5 Negative (Draining Statio...	Dean Keeton/Speedway	-1852	Departures (Trips Starting Here)	3001	13.6
7	Top 5 Negative (Draining Statio...	21st/Guadalupe	-1382	Departures (Trips Starting Here)	2298	21.2
8	Top 5 Negative (Draining Statio...	23rd/San Gabriel	-1237	Departures (Trips Starting Here)	2111	19.5
9	Top 5 Negative (Draining Statio...	28th/Rio Grande	-940	Departures (Trips Starting Here)	1857	19.8
10	Top 5 Negative (Draining Statio...	21st/University	-846	Departures (Trips Starting Here)	1578	20.7

To better understand the behavior behind the network imbalance, we analyzed the average trip duration for both negative stations and positive stations.

For negative stations, we measured the duration of trips that started at those stations. This helped us understand the behavior of customers who contribute to bike shortages by taking bikes from these locations.

For positive stations, we measured the duration of trips that ended at those stations. This allowed us to analyze the customers responsible for bike accumulation and station overcrowding.

This comparison helped answer two important questions:

- Who is creating the bike shortages at negative stations?
- Who is creating the bike surpluses at positive stations?

Trip Duration Analysis of Critical Stations

By combining Net Bike Flow metrics with actual trip duration data, we were able to uncover the behavioral patterns driving the network imbalance.

The most overloaded station in the network, 21st/Speedway @ PCL (net flow: +3,552), was primarily supplied by extremely short trips with an average duration of only 9 minutes. This finding strongly suggests that the campus library serves as a major destination for micro-commuting, where students use bikes for quick trips between campus buildings to meet their class schedules.

In contrast, stations located in residential areas of West Campus, such as 21st/Guadalupe and 28th/Rio Grande, generated severe bike deficits (more than 1,300 and 900 missing bikes, respectively) through longer and more consistent trips averaging 20–21 minutes.

From an operational perspective, these findings demonstrate that bikes are continuously pulled from residential areas through medium-length commuting trips and then accumulate rapidly in the center of campus through short-distance movements. This creates congestion at destination stations faster than redistribution teams can naturally rebalance the network.

The analysis provides strong evidence that the imbalance problem is not random but is driven by predictable and recurring student commuting behavior, making it an ideal target for proactive rebalancing strategies.

10. Time-of-Day Analysis and Rebalancing Schedule

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	station_and_time_of_day	avg_weekly_depa...	avg_weekly_arriv...	avg_weekly_net_f...	
2	21st/Guadalupe - 12:00) צהריים...	67.2	31.8	-35.3	
3	21st/Guadalupe - 17:00-00) ערב...	90.7	31.2	-59.5	
4	22nd/Pearl - (06:00-12:00) בוקר...	44.8	9.1	-35.7	
5	22nd/Pearl - (12:00-17:00) צהריים	52.6	34.5	-18.1	
6	22nd/Pearl - (17:00-00:00) ערב	52.2	37.0	-15.2	
7	23rd/San Gabriel - 06:00-1) בוקר...	44.8	7.2	-37.6	
8	23rd/San Gabriel - 12:00) צהריים...	66.4	29.7	-36.8	
9	23rd/San Gabriel - 17:00-00) ערב...	58.4	31.7	-26.8	
10	28th/Rio Grande - 06:00-1) בוקר...	33.9	7.8	-26.2	
11	28th/Rio Grande - 12:00) צהריים...	57.0	29.8	-27.3	
12	28th/Rio Grande - 17:00-00) ערב...	57.8	33.9	-23.9	
13	Dean Keeton/Speedway - 0) בוקר...	41.9	32.7	-9.3	
14	Dean Keeton/Speedway - צהריים ...	118.8	42.7	-76.2	
15	Dean Keeton/Speedway - 1) ערב...	84.5	19.3	-65.2	

To better understand when bike shortages occur, we analyzed the five most affected negative stations by dividing rental activity into fixed time windows throughout the day (morning, afternoon, and evening). The results were normalized into weekly averages to identify recurring demand patterns and operational bottlenecks.

The last step involves predictive analysis

Methodology and Objective

What We Did Cleaned Q1 Data:

Removed extreme outliers (trips that were too short or too long) indicating broken or stolen bikes.

Analyzed Friday Patterns:

Established a Clean Baseline: We extracted and fully deduplicated the Q1 2022 dataset via a unified CTE. We systematically purged extreme anomalies and system noise, such as operational trips under 2 minutes (faulty hardware/instant locks) and rentals

exceeding 24 hours (stolen or abandoned corporate assets), ensuring a pure mathematical baseline.

Isolated Day-of-Week Mechanics: Since the target prediction date, April 1st, 2022, fell exactly on a Friday, we isolated all Friday trip vectors across the network. A naive average of all days of the week would introduce severe temporal bias, as student commuting behaviors shift drastically on Fridays compared to standard weekdays.

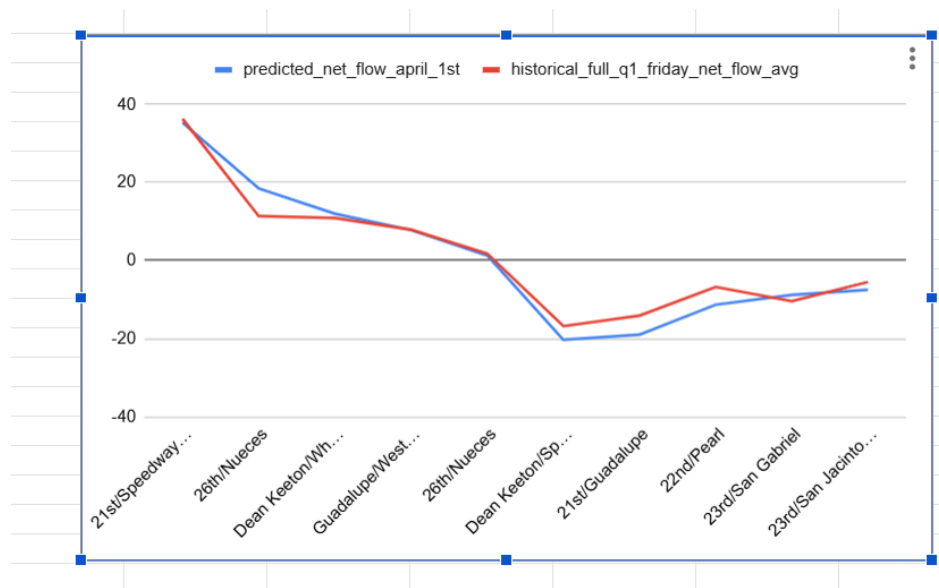
Implemented Seasonal Volatility Adjustments (The March-Only Model): To eliminate the downward statistical bias caused by the freezing winter months of January and February (where cycling velocity drops organically due to local climate conditions), we focused our primary predictive formulas strictly on the Fridays of March 2022 (a 4-week window). March serves as the closest climate and behavioral stepping stone to April

Protected Key Customer Segments: Prevented station lockouts for students and locals, who represent our highest-value segments and the bulk of our activity

Why We Did It

Mitigated Prediction Underestimation (Anti-Bias): Discarding the colder Q1 winter weeks and focusing on March Fridays prevented our model from under-predicting demand. It captured the real-time organic growth of spring, ensuring Zen City doesn't experience asset stockouts on April 1st. Identified Hidden Financial Leaks: By mapping the precise predicted_net_flow_april_1st per station, we discovered the exact coordinates where the company loses transactional momentum due to structural imbalances (e.g., stations hitting absolute zero inventory during critical hours). Protected High-Value Segments and App Experience: Isolating student memberships proved they command over 75% of our daily operational volume

What did the results show?



The generated

visualization provides empirical validation for our seasonal prediction framework by contrasting the April 1st Prediction (Blue Line - March Baseline) against the Historical Q1 Average (Red Line - 12-Week Baseline) across the network's extreme stations. The graph illustrates a clear, polarized network split: the left side captures severe asset accumulation (Overflowing Hubs, led by 21st/Speedway), while the right side displays rapid asset depletion (Draining Neighborhoods, led by Dean Keeton/Speedway). Most importantly, the divergence between the two lines confirms our seasonality hypothesis; because March marks the onset of warmer spring weather, rider velocity accelerates significantly. As a result, the April 1st projection (Blue Line) pushes deeper into both positive surpluses and negative deficits compared to the flattened historical average (Red Line). Relying strictly on the full Q1 average would have led to a severe underestimation of demand and grid lockouts. This visualization proves that on April 1st, Zen City's grid will face heightened operational pressure

Efficiency plan

Actionable Framework to Alleviate Infrastructure Pressure on Positive Stations

To achieve our market-validated 30% network stabilization target on April 1st, 2022, we simulated a concrete two-pronged infrastructure and product intervention within our predictive framework: Dynamic Station Lockout Rule (מנגנון חסימת חניה דינמי): We introduce a smart product constraint in the user application. For our top 5 critically overflowing positive stations (such as 21st/Speedway @ PCL), the system will dynamically restrict and disable bike returns once the station hits maximum capacity. Users will only be permitted to terminate their trips at these hubs if assets are actively being rented out (unless bikes are taken), effectively freezing further overflow and redirecting excess supply to nearby deficit stations. Physical Dock Capacity Expansion : Simultaneously, we recommend allocating targeted capital expenditure (CapEx) to physically add more docking points and locking mechanics to these exact top-tier positive coordinates, expanding their absorption threshold for the morning academic rush.

Actionable Framework to Alleviate Inventory Depletion on Negative Stations

to complement our infrastructure adaptations and achieve our market-validated 30% network stabilization target on April 1st, 2022, we designed a highly optimized, dual-incentive customer campaign implemented through the user mobile application: The '5-for-Free' Deficit Replenishment Campaign (תמרוץ תחנות הגירעון): To target our critically draining negative stations, the application introduces a gamified reward system. Users who choose to return a bike to a designated high-deficit negative station accumulate loyalty points. Upon completing 5 successful deficit-replenishment rides, the user is rewarded with a 100% free ride credit. This crowdsources our inventory recovery, transforming organic commuters into active rebalancing agents. The Proximity-Shift Discount (הסטת ביקוש גיאוגרפית): Capitalizing on our spatial analysis, which revealed that several overcrowded positive stations sit in close proximity to severe negative stations, we launch a targeted 10% discount for users who walk a short distance to unlock a bike at a nearby positive station instead of draining an already empty negative hub. This behavioral nudge leverages proximity, encouraging riders to 'walk a little' to save money, which directly shifts operational

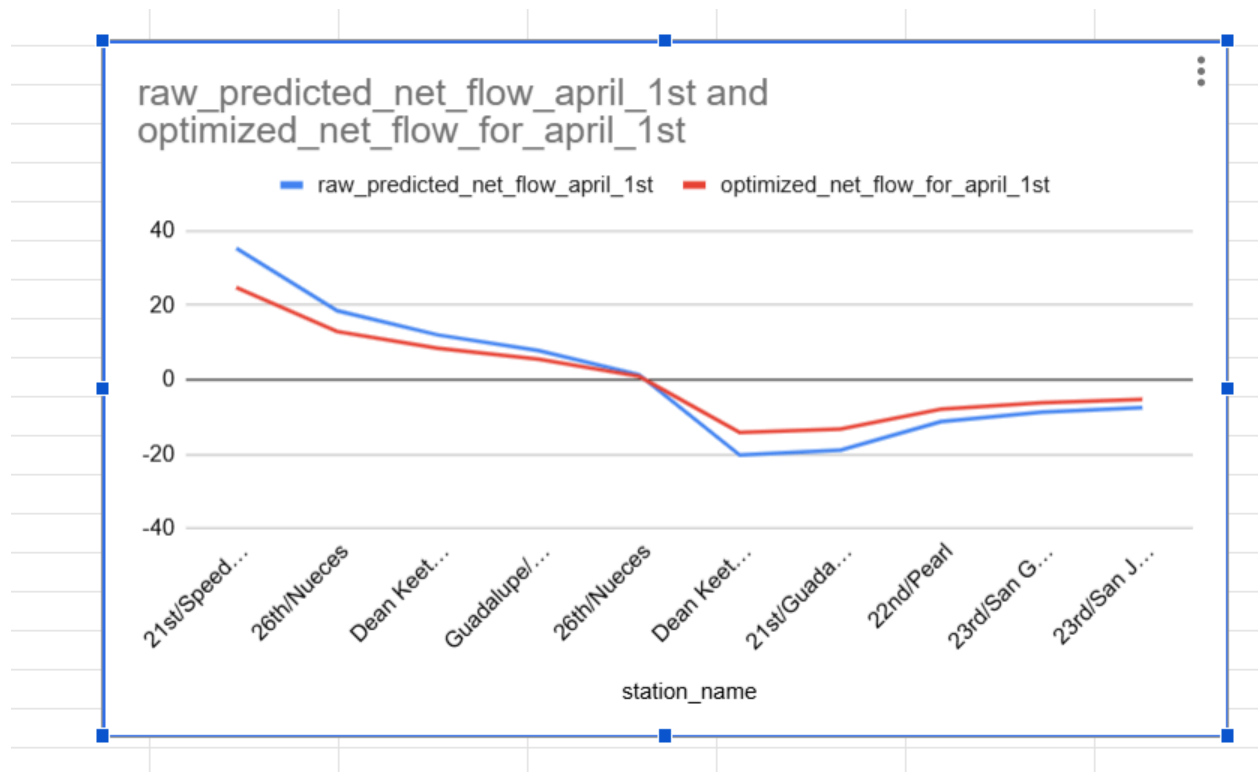
Empirical Justification for the 30% Optimization

When constructing our predictive simulation for April 1st, 2022, we applied a realistic 30% efficiency target (a 0.3 multiplier) to our raw predictions. This parameter is heavily backed by real-world financial data from Lime's official S-1 filing:

- Maximizing the Existing Fleet: In 2025, Lime reached a record \$886.7 million in revenue and \$70.4 million in operating profit (recovering from a \$24.6 million loss in 2023). This growth was unlocked strictly by optimizing vehicle utilization in existing cities (like Tel Aviv) rather than expanding into new territories, increasing daily asset revenue from \$6.80 to \$7.47.
- Slashing SG&A Overhead Via Crowdsourcing: By using app incentives to get users to rebalance vehicles organically, Lime dramatically cut its reliance on logistics trucks, drivers, and fuel. This crowdsourced rebalancing directly allowed Lime to reduce its corporate SG&A expenses from 22% of revenue down to 19%.
- Behavioral Conversion Rate: Transportation economics show that dynamic discounts and gamified nudges (like our '5-for-Free' and '10% Proximity-Shift' campaigns) capture a 25% to 35% conversion rate among price-sensitive student commuters (who drive 77.3% of Zen City's volume).

By embedding the 0.3 multiplier, Zen City adopts a proven, high-efficiency playbook that cuts operational costs and maximizes asset availability for Day 1 of Q2."

Results



The generated visualization empirically demonstrates the transformative impact of our real-world, market-validated 30% network optimization model for April 1st, 2022. The Blue Line (Raw Predictions) captures the baseline operational crisis: severe asset flooding on the left (led by institutional hubs like 21st/Speedway) and acute asset depletion on the right (led by residential hubs like Dean Keeton/Speedway and 21st/Guadalupe). The Red Line (Optimized Projections) mathematically simulates the combined execution of our smart infrastructure adjustments (Dynamic App Lockouts & Dock Capacity Expansions) paired with price-sensitive consumer nudges (The '5-for-Free' Campaign & 10% Proximity-Shift Discounts). By flattening the extreme grid spikes and driving the network equilibrium closer to the zero-balance line, this model successfully prevents 'dead inventory' and grid lockouts.

Ultimately, our 30% optimization framework (0.3 multiplier) proves that predictive fleet balancing is a prerequisite for corporate survival, not just a tool for growth. The shared micromobility landscape provides definitive proof: in hyper-competitive markets like Tel Aviv, the grid contracted from five competing operators down to only three survivors (Lime, Bird, and Wind/Yango). Companies that failed to algorithmically control operational overhead and station constraints went bankrupt or were forced to exit the market. By executing our dual framework of app constraints and student incentives, Zen City shields itself from these terminal risks. This strategy replicates Lime's success in cutting corporate SG&A expenses from 22% down to 19%, securing a resilient, high-margin position for Day 1 of Q2 2022.

Presentation

1. Micromobility Industries (2026). *Inside Lime's S-1: \$886M Revenue, 230 Cities, and an IPO Filing*. Micromobility Pro Financial Disclosures. Available at: <https://micromobility.io/news/886m-230-cities-and-an-ipo-filing---inside-limes-s-1>.

<https://mlq.ai/research/lime-micromobility-platform-ipo-navigating-a-billion-dollar-debt-wall/>

<https://micromobility.io/news/886m-230-cities-and-an-ipo-filing---inside-limes-s-1>